KisanKhata

Product Requirements Document

Simple farm expense management and farm-fresh marketplace for farmers

Version	1.0 (Draft for stakeholder review)
Date	11 October 2026
Product type	Web application (HTML frontend, Java backend)
Status	In review
Previous working name	TechFarm (renamed to KisanKhata)

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1. Document Control

Field	Value
Product name	KisanKhata
PRD title	KisanKhata - Product Requirements Document
Version	1.0
Status	Draft - ready for stakeholder review
Document owner	Product Owner
Date	11 October 2026
Intended audience	Product, Engineering, Design, QA, Operations, Business stakeholders, Evaluators

Revision History

Version	Date	Author	Summary of Changes
0.1	11 Oct 2026	Product Team	Initial draft derived from the TechFarm project report
1.0	11 Oct 2026	Product Team	Full PRD: renamed to KisanKhata, scope fixed to original features only, simplified UI and single-JAR deployment added

Stakeholders and Approvers

Role	Responsibility	Sign-off
Product Manager / Owner	Scope, priorities, acceptance of the PRD	Pending
Engineering Lead (Java)	Feasibility, backend design, deployment package	Pending
Design Lead	Simple, farmer-friendly UI in plain HTML/CSS	Pending
QA Lead	Test plan from acceptance criteria	Pending
Operations Lead	Hosting, backup, support of the running system	Pending
Business / Founder	Pilot goals, marketplace operation, UPI account	Pending
Domain advisor (farmer representative)	Validates wording, flows and usability for farmers	Pending

How to Read Requirement Priorities

Priority	Meaning
P0	Required for MVP / launch. Product cannot be released without it.
P1	Important, fast-follow after launch (or soft-launch checkpoint) if time allows.
P2	Desirable, future phase. Not planned for the current release.

Requirement IDs follow the pattern **FR-MODULE-NN** (functional), **NFR-NN** (non-functional), **BG/PG** (business/product goals) and **KPI-NN** (metrics). Assumptions are labelled **A-NN** and open questions **Q-NN**.

2. Introduction and Product Vision

2.1 Purpose

This PRD defines what KisanKhata must do, for whom, and how each requirement will be accepted. It is the single reference for engineering, design, QA and stakeholders. It is derived from the original TechFarm college project report and deliberately keeps the same feature set, rewritten as a clear product specification. No new features beyond that report have been added; the only additions are the minimum needed to make the existing features work safely (these are listed as assumptions).

2.2 Product Vision

Every farmer should be able to see, in one simple place, what each crop costs, who was paid, what work was done and what is coming next - without paperwork, without calculators and without needing technical skills.

2.3 Product Summary

KisanKhata is a very simple website where farmers record crops, expenses, workers, water and fertilizer/pesticide use, set reminders and view expense reports, and where buyers can order products from an online mart run by an admin.

2.4 Product Scope

This PRD covers three user areas of one website:

- **Farmer Management** - registration, login, crops, expenses, workers, water, fertilizer/pesticide, reminders, reports.
- **KisanKhata Mart** - product browsing, cart, checkout, UPI payment instructions and bill for buyers.
- **Admin** - product management, user list, orders and sales reports, payment settings.

This PRD does not cover: source code, low-level implementation, detailed API contracts, pixel-perfect UI design and database schema implementation. Technology choices are limited to the constraints in Section 8.4.

3. Problem Statement and Market Context

3.1 The Problem

- **Farmer:** expenses, worker details and daily work are kept in memory or on paper. Calculations take time and mistakes are common, so the farmer often does not know the true cost of a crop and ends up with less profit than expected.
- **Buyer:** there is no simple, direct place to buy farm products (for example grapes) from the farm operator.
- **Admin / shop operator:** product details, orders and payments are tracked manually.

3.2 Current Problems / Existing Alternatives

Alternative	How it is used today	Limitation
Paper notebook / diary	Expenses and worker days written by hand	Records get jumbled or lost; manual totals; no reports
Memory	Remembering workers and daily schedule	Unreliable; details forgotten
Phone calculator / messaging apps	Quick totals, reminders to self	No history per crop or per period
General spreadsheets	Some farmers use Excel	Needs computer skills; not designed for farm tasks
Phone calls / local traders	Selling or buying produce	No catalogue, no bill, no record

3.3 Pain Points

- Difficult to remember details of each worker and what each was paid.
- Manual calculation is slow and error-prone; paperwork is heavy.
- Records are hard to keep tidy; reports by date or by crop are not available.
- Daily schedule (watering, spraying) is easily forgotten.
- Business: no data to plan total expenditure before the season.

3.4 Why Now

Smartphones and mobile internet are widely used by farmers, and many already use UPI for payments. A very simple web tool can therefore replace the notebook. **No market statistics are quoted here.** Market validation (interviews with farmers, a pilot with a small group) is required and is listed in Open Questions.

3.5 Competitive Landscape

No competitor research has been provided, so no competitor facts are stated. Categories to be reviewed during validation:

Category	Expected table-stakes	Potential differentiation for KisanKhata
Generic farm / agriculture apps	Crop and expense records, reminders	Extremely simple screens, big buttons, no extra features
Bookkeeping / expense tracker apps	Expense entry, reports	Crop-wise and worker-wise views built for farm use
Online produce marketplaces	Catalogue, cart, payment	Small, direct mart connected to the same farm website

3.6 Product Differentiation

- **Simplicity first:** few screens, large buttons, plain words, no clutter.
- **Everything in one place:** farm records and the mart under one website.
- **Easy to run:** one Java file to start; no database or web server to install.

4. Goals, Objectives and Non-Goals

4.1 Business Goals

Targets below are **proposed pilot targets** to be confirmed by the business; they are not based on market data.

ID	Goal	Target	Timeframe
BG-01	Farmers adopt the farm management module	50 registered farmers in the pilot	3 months after launch
BG-02	Farmers keep using it for real records	30 farmers with at least 1 expense entry every week of a month	3 months after launch
BG-03	Marketplace completes real orders	20 orders marked Paid	3 months after launch
BG-04	Deployment is easy	New instance running in 30 minutes or less by a non-expert following one page of steps	Before launch
BG-05	Low running cost	Runs on one low-cost server with no paid third-party service	At launch

4.2 Product Goals

ID	Goal	Linked Business Goal
PG-01	Replace the paper notebook for expenses, workers and daily farm work	BG-01, BG-02
PG-02	Let a farmer see total cost per crop and per period	BG-02
PG-03	Let a farmer see what each worker is owed	BG-02
PG-04	Make every screen usable by a first-time smartphone user	BG-01, BG-02
PG-05	Allow buyers to order products and pay via UPI with a clear bill	BG-03
PG-06	Make the product deployable and maintainable by one person	BG-04, BG-05

4.3 Non-Goals

- No advanced analytics, profit prediction, weather, market-price feeds or AI features.
- No mobile app (the website works in a phone browser).
- No online payment gateway; the mart only shows UPI payment details and the admin confirms payment.
- No SMS, email or push notifications; reminders are shown inside the website.
- No multi-seller marketplace; products are added only by the admin.
- No delivery tracking, discounts, coupons, ratings or reviews.
- No multi-language support in release 1 (text is kept ready for translation).
- No illegal-chemical or market information service (mentioned as a problem in the original report; not a feature here).

5. Success Metrics and KPIs

5.1 North Star Metric

Weekly Active Farmers (WAF) - number of farmers who save at least one record (expense, worker, water, fertilizer/pesticide or reminder) in a calendar week. It reflects the core value: farmers keep real records in the system instead of on paper.

5.2 Primary KPIs

KPI	Definition	Target (proposed)	Measurement Method
KPI-01 Registrations	Farmer accounts created	50 in 3 months	Count of farmer_registered events
KPI-02 Activation	% of new farmers who add 1 crop and 1 expense within 7 days of registering	60%	Event funnel (Section 17)
KPI-03 Weekly Active Farmers	See North Star	30 by month 3	Weekly query of saved records
KPI-04 Retention	% of activated farmers still active in week 4	40%	Cohort by registration week
KPI-05 Report usage	% of active farmers opening any report per month	50%	report_viewed events
KPI-06 Mart conversion	Orders placed / buyers who added to cart	30%	cart_item_added vs order_placed
KPI-07 Orders completed	Orders marked Paid	20 in 3 months	Admin order report
KPI-08 Task speed	Median time to add an expense after login	30 seconds or less	Usability test with 5 farmers
KPI-09 Deployment time	Time to run a new instance	30 minutes or less	Timed test by a non-expert

5.3 AARRR Framework

Stage	Definition for KisanKhata
Acquisition	Farmer or buyer opens the home page and starts registration (visits and registrations)
Activation	Farmer adds first crop and first expense; buyer adds first product to cart
Retention	Farmer saves records in consecutive weeks; buyer places a repeat order
Revenue	Value of orders marked Paid in the mart (no subscription in release 1)
Referral	Not measured in release 1 (no referral feature)

5.4 Guardrail Metrics

Guardrail	Limit
Application error rate (server errors per 100 requests)	Below 1
Failed saves caused by validation confusion (form submitted with error then abandoned)	Below 15% of form submits
Page response time (95th percentile)	Below 1 second for normal pages
Orders stuck in Payment Reported for more than 2 days	Zero
Security incidents (unauthorised data access)	Zero
Data loss events	Zero

6. Target Users and Personas

Personas are fictional and represent the expected users. They must be checked against real farmer interviews.

Persona A - Primary User: Ramesh, Grape Farmer

Attribute	Description
Age / role	45, owns a 4-acre vineyard, employs 3 to 6 workers in season
Context	Keeps expenses in a notebook; calculates totals at season end; forgets spray and watering dates
Goals	Know crop cost; remember worker days and wages; see reminders
Frustrations	Paperwork, mistakes in totals, small text, long forms, English-only technical words
Needs	Big buttons, few fields, clear confirmation after each save, simple words
Behaviour	Uses WhatsApp and UPI daily; types slowly; prefers pick-lists and date pickers
Device / environment	Android phone with Chrome, mobile data, sometimes weak network; outdoors, bright light
Frequency	Several times a week in season; once a week otherwise

Persona B - Farmer Family Member: Sunita, Farm Assistant

Attribute	Description
Age / role	28, helps with farm accounts
Goals	Enter expenses and worker details for the family; print or read reports
Frustrations	Re-entering the same information; unclear error messages
Device	Shared phone or home laptop; Chrome
Frequency	Weekly

Persona C - Buyer: Priya, Customer

Attribute	Description
Age / role	32, buys fresh farm products for her family
Goals	See products and prices, order quickly, pay by UPI, get a bill
Frustrations	Forced long sign-up, unclear total, uncertainty whether payment was received
Device	Phone, UPI app installed
Frequency	Occasional

Persona D - Admin: Mr. Patil, Mart Operator

Attribute	Description
Role	Runs the mart and the website
Goals	Add products, keep stock correct, check payments, see orders and users
Frustrations	Manual order lists; mismatched payments
Device	Laptop or phone; checks daily

7. User Stories and Jobs To Be Done

Farmer

- As a farmer, I want to register and log in with my email, so that my records are private to me.
- As a farmer, I want to add my crops with planting date and duration, so that I know what I am growing.
- As a farmer, I want to add an expense against a crop, so that I know how much each crop costs.
- As a farmer, I want to correct or delete a wrong expense, so that my totals are right.
- As a farmer, I want to save worker details and days worked, so that I know what to pay each worker.
- As a farmer, I want to note when I watered and when I used fertilizer or pesticide, so that I do not repeat or forget.
- As a farmer, I want to set a reminder, so that I remember tasks on the right day.
- As a farmer, I want reports by date, by crop and by worker, so that I can see my spending clearly.

Buyer

- As a buyer, I want to see products and prices without logging in, so that I can decide quickly.
- As a buyer, I want to add products to a cart and change quantity, so that I order what I need.
- As a buyer, I want to check out with my delivery address and pay by UPI, so that my order is confirmed.
- As a buyer, I want a bill with a number, so that I have proof of my order.

Admin

- As an admin, I want to add, edit and delete products with price and stock, so that the mart is accurate.
- As an admin, I want to see all registered farmers and buyers, so that I know who uses the system.
- As an admin, I want to see orders and mark them paid after checking my UPI app, so that payments are tracked.
- As an admin, I want to set my UPI details, so that buyers pay the right account.

Jobs To Be Done

Job	When	So I can
Keep track of what I spend on each crop	Every time I pay for something	Know my real profit or loss
Pay workers correctly	At week or season end	Avoid disputes
Remember farm tasks	Before important dates	Protect crop health
Sell products to customers directly	When harvest or stock is available	Earn more with less effort

8. Scope, Assumptions, Constraints and Dependencies

8.1 In Scope

- Public home, About and Contact pages.
- Farmer: registration, login/logout, crops, expenses, workers (with payment amount), water records, fertilizer/pesticide records, reminders, three reports, help page.
- Mart: browse, register/login, cart, checkout, UPI payment page, bill, my orders.
- Admin: login, product add/edit/delete, user list, orders report, mark paid, payment settings.
- Single-JAR deployment package with one-page setup instructions and backup guidance.

8.2 Out of Scope

- Everything listed in Non-Goals (Section 4.3).
- Payment gateway integration, refunds processing, delivery logistics.
- Farmer-to-buyer selling by farmers themselves.
- Native mobile apps, offline mode, data import/export tools.

8.3 Assumptions

ID	Assumption (to be confirmed)
A-01	The mart is operated by a single seller (the admin). Farmers do not list products.
A-02	Farmer accounts and buyer accounts are separate logins, as in the original design. One email can exist in each.
A-03	The first admin account is created from the setup configuration file at first start. There is no public admin sign-up (the original report had one; removed for safety).
A-04	Payment is by static UPI QR / UPI ID set by the admin. The buyer pays in their own UPI app, taps "I have paid", and the admin verifies in their UPI app and marks the order Paid.
A-05	Workers need a daily wage field so that the Payment Amount column (empty in the original screens) can be calculated as days worked x daily wage.
A-06	Reminders are shown inside the website only (home banner and list). No SMS or email.
A-07	Data is stored in an embedded, file-based database inside the application folder so that nothing else must be installed. Final engine to be chosen by Engineering (Q-02).
A-08	Currency is Indian Rupee; dates are shown as dd-mm-yyyy; UI language is English with simple words.
A-09	A crop that has expenses cannot be deleted (prevents silent loss of expense history).
A-10	Product deletion hides the product from the mart; past orders keep the product name and price as ordered.
A-11	Stock quantity is tracked; it reduces when an order is placed and returns if the order is cancelled.
A-12	Expected load in the first year is small (under 100 simultaneous users).

8.4 Constraints

Type	Constraint
Technology - frontend	Plain HTML, CSS and light JavaScript only. No frontend framework and no build step.
Technology - backend	Java only (Java 17 or later, long-term-support version). No other backend language.
Technology - database	Embedded database, no separate database server to install (see A-07).
Deployment	One runnable JAR containing the server and the HTML pages; started with a single command; one configuration file.
Business	No paid third-party services in release 1.
Operational	One person should be able to run, back up and restart the system.
Geographic	India: INR, UPI, 10-digit mobile numbers, 6-digit PIN codes.
Compliance	Personal data (name, phone, address, email) is collected; a simple privacy notice and password protection are required. Specific legal requirements must be confirmed with a qualified advisor (Q-10); none are assumed here.

Type	Constraint
Budget / timeline	Not provided. Roadmap in Section 19 is sequenced by priority, not by dates.

8.5 Dependencies

Dependency	Type	Description	Impact if Delayed
Java runtime (17+) on host	Technical	Required to run the JAR	Cannot deploy
Host / server with internet access	Operational	Any machine or low-cost cloud server that can run Java	No public access
Admin UPI ID and QR image	Business	Needed for the payment page	Mart cannot take payments
Product catalogue and photos	Business	Initial products and images supplied by admin	Mart empty at launch
Farmer feedback sessions	Product	Validate wording and flow with real farmers	Risk of poor usability
Domain name and HTTPS certificate	Operational	Needed for secure public access	Passwords sent unencrypted if skipped

9. Functional Requirements

Every requirement states WHAT the system must do and how it is accepted. Validation messages must be short, in simple words and shown next to the field. After every successful save the screen shows a clear green "Saved" message.

9.1 Public Pages and Farmer Home

ID	Requirement	Priority	Acceptance Criteria
FR-HOME-01	Visitor sees a home page with the product name and two entry areas: Farmer Management (Login, Sign Up) and KisanKhata Mart (Login, Sign Up, Visit Mart), plus links to About and Contact.	P0	Each button opens the correct page. Page is usable at 360 px width with no sideways scrolling.
FR-HOME-02	Visitor can open static About and Contact pages.	P1	Both pages open from the home page and show text and contact details set in the configuration file.
FR-HOME-03	After login, Farmer sees Farmer Home with greeting by name, today's date and large buttons: Crops, Expenses, Workers, Water, Fertilizer/Pesticide, Reminders, Reports, Help.	P0	All eight buttons are visible and each opens its module. On a phone they appear in one column with buttons at least 48 px high.
FR-HOME-04	Farmer can open a Help page with plain-language steps for each module.	P1	Help page shows at least one short step list per module.

9.2 Farmer Authentication

ID	Requirement	Priority	Acceptance Criteria
FR-AUTH-01	Visitor can register as a Farmer with first name, last name, mobile number, address, email, password and repeat password.	P0	Valid data creates the account and opens the login page with "Registration successful". Any empty required field shows a message and nothing is saved. Email already used shows "This email is already registered". Mobile not 10 digits, password under 6 characters, or passwords not matching are rejected with a message.
FR-AUTH-02	Farmer can log in with email and password.	P0	Correct details open Farmer Home. Wrong details show one message "Email or password is incorrect" (does not say which). Empty fields show a message. A disabled account shows "Account is disabled. Contact admin".
FR-AUTH-03	Farmer can log out from every page.	P0	A Log out button is visible in the header of every Farmer page. After logout the Back button does not show Farmer data; protected pages redirect to login.
FR-AUTH-04	System ends a Farmer session after 30 minutes without activity.	P0	First request after 30 minutes opens login with "Session expired, please log in again". Activity within 30 minutes keeps the session.
FR-AUTH-05	System allows a Farmer to see and change only their own data.	P0	Opening a Farmer page without login redirects to login. Requesting another farmer's record (for example by editing an ID in the address) returns "Not allowed" and no data.
FR-AUTH-06	System temporarily blocks repeated failed logins.	P1	After 5 failed logins for one email within 10 minutes, further attempts are refused for 10 minutes with a clear message. Login works again after the wait.

9.3 Crop Management

ID	Requirement	Priority	Acceptance Criteria
FR-CROP-01	Farmer can add a crop with name, category, plantation date, plantation duration (days) and description.	P0	Name, category, date and duration are required. Date is a valid date; duration is a whole number from 1 to 999. Saved crop appears in the list at once. Invalid input saves nothing and shows a message.
FR-CROP-02	Farmer can view their crops in a list showing name, category, plantation date, duration and description.	P0	Only the farmer's own crops are listed. With no crops the page shows "No crops added yet. Add your first crop."

ID	Requirement	Priority	Acceptance Criteria
FR-CROP-03	Farmer can edit a crop.	P1	Edited values are saved and shown in the list and reports; same validation as adding.
FR-CROP-04	Farmer can delete a crop after confirming.	P0	Confirmation "Delete this crop?" appears; Cancel keeps it. A crop with expenses is not deleted and shows "This crop has expenses. Delete those expenses first." (A-09).
FR-CROP-05	System rejects a duplicate crop (same name and plantation date) for the same farmer.	P1	Second identical crop is refused with "This crop is already added".

9.4 Expense Management

ID	Requirement	Priority	Acceptance Criteria
FR-EXP-01	Farmer can add an expense with details, date, crop and amount.	P0	All fields required. Crop must be one of the farmer's crops (pick-list). Amount greater than 0, up to 2 decimals, at most 9,999,999. Date cannot be in the future. If the farmer has no crop, the form shows "Add a crop first" with a link.
FR-EXP-02	Farmer can view their expenses with date, details, crop and amount, and a total.	P0	List shows only own expenses, newest first, with correct total. Empty state: "No expenses yet."
FR-EXP-03	Farmer can edit an expense.	P0	Saved changes appear in the list and all reports. Same validation as adding.
FR-EXP-04	Farmer can delete an expense after confirming.	P0	Confirmation appears; after confirming the expense disappears from list and reports.
FR-EXP-05	System creates only one record when Add is pressed twice quickly.	P0	Double click or double tap on Add creates exactly one expense; button is disabled after the first press until the response arrives.

9.5 Worker Management

ID	Requirement	Priority	Acceptance Criteria
FR-WRK-01	Farmer can add a worker with name, contact number, days worked, daily wage, and optionally email, address and UPI ID.	P0	Name, contact (10 digits), days worked (0 to 366) and daily wage (greater than 0) are required. Optional email, if given, must be a valid email. Saved worker appears in the list.
FR-WRK-02	Farmer can view their workers in a list with days worked and payment amount.	P0	Only own workers shown. Empty state: "No workers added yet."
FR-WRK-03	Farmer can edit a worker (for example update days worked).	P0	Changes are saved and payment amount is recalculated.
FR-WRK-04	Farmer can delete a worker after confirming.	P0	Confirmation appears; deleted worker disappears from list and worker report.
FR-WRK-05	System calculates each worker's payment amount as days worked x daily wage.	P0	2 days at Rs. 500 shows Rs. 1000. Changing days or wage changes the amount shown.

9.6 Water and Fertilizer/Pesticide Records

ID	Requirement	Priority	Acceptance Criteria
FR-WTR-01	Farmer can record watering with plot name or number, with fertilizer (Yes/No), quantity with unit (Litres or Hours) and date.	P0	All fields required. Quantity greater than 0. Date not in the future. Saved record appears in the water list.
FR-WTR-02	Farmer can view water records in a list.	P0	Only own records, newest first. Empty state shown when none.
FR-WTR-03	Farmer can delete a water record after confirming.	P0	Record disappears after confirming; Cancel keeps it.
FR-PST-01	Farmer can record fertilizer or pesticide use with plot name or number, product name, quantity with unit (ml or kg) and date.	P0	All fields required. Quantity greater than 0. Date not in the future. Saved record appears in the list.
FR-PST-02	Farmer can view fertilizer/pesticide records in a list.	P0	Only own records, newest first.

ID	Requirement	Priority	Acceptance Criteria
FR-PST-03	Farmer can delete a fertilizer/pesticide record after confirming.	P0	Record disappears after confirming.

9.7 Reminders

ID	Requirement	Priority	Acceptance Criteria
FR-REM-01	Farmer can set a reminder with time, date and description.	P0	All fields required. Description up to 100 characters. A date and time in the past is rejected with "Choose a future date and time".
FR-REM-02	Farmer can view reminders sorted by date and time.	P0	Upcoming reminders are listed first; past ones are shown as Done/Past.
FR-REM-03	Farmer can delete a reminder after confirming.	P0	Reminder disappears after confirming.
FR-REM-04	Farmer Home shows a highlighted banner for reminders due today and any overdue reminders.	P0	A reminder dated today appears in the banner at login and on each visit to Farmer Home. Reminders with no entry for today show no banner.

9.8 Reports

ID	Requirement	Priority	Acceptance Criteria
FR-RPT-01	Farmer can view a date-wise expense report for a From date and To date.	P0	Lists expenses in the range (both dates included) with total. From date after To date shows an error. No data shows "No expenses in this period."
FR-RPT-02	Farmer can view a crop-wise expense report for a selected crop.	P0	Lists all expenses of that crop with total. Crop with no expenses shows an empty message.
FR-RPT-03	Farmer can view a worker-wise payment report with name, contact, days worked, payment amount and grand total.	P0	Amounts match Section 9.5 calculation; grand total equals the sum of rows.
FR-RPT-04	Farmer can open a print-friendly version of any report.	P1	Print view hides menus and buttons and fits an A4 page width.

9.9 Mart - Buyer

ID	Requirement	Priority	Acceptance Criteria
FR-MART-01	Visitor can browse products without logging in, seeing image, name, price and availability.	P0	Only active products are shown. Zero stock shows "Out of stock".
FR-MART-02	Visitor can register as a Buyer with first name, last name, mobile number, shipping address, email and password.	P0	Same validation rules as FR-AUTH-01. Buyer accounts are separate from Farmer accounts (A-02).
FR-MART-03	Buyer can log in and log out.	P0	Same behaviour as FR-AUTH-02 to FR-AUTH-04.
FR-MART-04	Buyer can add a product to the cart; a visitor who is not logged in is asked to log in first.	P0	After login the visitor returns to the product list. Adding the same product again increases its quantity. Out-of-stock products cannot be added.
FR-MART-05	Buyer can view the cart with items, price, quantity, line total and grand total; change quantity; remove items; continue shopping.	P0	Quantity is a whole number from 1 up to available stock. Totals update correctly. Empty cart shows "Your cart is empty."
FR-MART-06	Buyer can check out by confirming delivery address, city, PIN code and contact number (pre-filled from profile).	P0	All fields required; PIN is 6 digits; contact is 10 digits. Placing the order creates a unique bill number with status Pending Payment, reduces stock and empties the cart.
FR-MART-07	System checks stock when the order is placed.	P0	If stock is not enough for any item, no order is created and the message names the product and available quantity.
FR-MART-08	Buyer sees a payment page with UPI QR/ID, order total and bill number, and buttons "I have paid" and "Cancel".	P0	"I have paid" sets status to Payment Reported and shows a thank-you message. "Cancel" returns to the order with status unchanged. The system never handles money.
FR-MART-09	Buyer sees a bill after placing an order with bill number, date, items, amount and status.	P0	Bill totals match cart totals at the time of ordering.

ID	Requirement	Priority	Acceptance Criteria
FR-MART-10	Buyer can view My Orders.	P1	Shows own orders only, newest first, with status.
FR-MART-11	Buyer can cancel an order while it is Pending Payment.	P1	Order becomes Cancelled and stock is returned.

9.10 Admin

ID	Requirement	Priority	Acceptance Criteria
FR-ADM-01	Admin can log in. The admin account is created from the setup configuration at first start; there is no public sign-up.	P0	Correct details open Admin Home. Wrong details show a generic message. Farmer or buyer credentials cannot open admin pages.
FR-ADM-02	Admin can log out; admin session ends after 30 minutes of inactivity.	P0	Same behaviour as FR-AUTH-03 and FR-AUTH-04.
FR-ADM-03	Admin can add a product with name, description, category, price, quantity and image.	P0	Required: name, price (greater than 0), quantity (whole number, 0 or more). Image JPG or PNG up to 1 MB. New product appears in the mart at once.
FR-ADM-04	Admin can view all products with Edit and Delete buttons.	P0	List shows name, price, stock and image thumbnail.
FR-ADM-05	Admin can edit a product.	P0	Changes show in the mart at once; same validation as adding.
FR-ADM-06	Admin can delete a product after confirming.	P0	Product disappears from the mart; past orders still show the name and price as ordered (A-10).
FR-ADM-07	Admin can view lists of farmers and buyers with name, email, mobile and registration date.	P0	Both lists load; passwords are never shown.
FR-ADM-08	Admin can deactivate and reactivate a user account.	P1	A deactivated user cannot log in (message in FR-AUTH-02); reactivation restores access. Data is kept.
FR-ADM-09	Admin can view an orders report filtered by date range and status, with total value.	P0	Rows show bill number, buyer, date, amount, status. Total equals sum of listed Paid orders.
FR-ADM-10	Admin can mark an order Paid or Cancelled.	P0	Only orders in Pending Payment or Payment Reported can be marked Paid. Each change records who and when. Cancelling returns stock.
FR-ADM-11	Admin can set the UPI ID, QR image and shop contact shown on the payment page.	P0	Saved values appear on the next payment page. Payment page shows a notice if not yet set.
FR-ADM-12	Admin Home shows counts of farmers, buyers, products and orders.	P1	Counts match the lists.

10. Edge Cases and Exception Handling

Feature	Scenario	Expected System Behaviour
Registration	Email already exists	Show "This email is already registered"; offer link to login; no new account
Registration	Two submits at once	Only one account created; second shows already-registered message
Login	Wrong password	Generic error; no hint which field was wrong
Login	Repeated failures	Temporary block (FR-AUTH-06) with clear wait message
Session	Expired session while filling a form	Redirect to login with message; unsaved form data is lost (user told)
Any Farmer page	Not logged in or another farmer's record requested	Redirect to login / "Not allowed"; no data returned
Add expense / crop / worker	Empty or invalid field	Message beside the field; nothing saved; other entered values kept
Add expense	No crops exist	"Add a crop first" with link
Delete crop	Crop has expenses	Deletion refused with explanation
Any save	Network drops or request times out	Message "Could not save. Please check internet and try again"; retry creates no duplicate
Any save	Server error	Friendly error page; error logged; data not half-saved
Reports	No data in range	Empty-state message, not a blank page
Reminder	Time in the past	Rejected with message
Cart	Quantity above stock	Quantity limited; message shows available stock
Cart	Product deleted or out of stock while in cart	Item flagged "No longer available" and excluded from total
Checkout	Two buyers order the last item	First order succeeds; second gets stock message; stock never negative
Payment	Buyer closes page before tapping "I have paid"	Order stays Pending Payment; buyer can return via My Orders
Payment	Buyer taps "I have paid" but did not pay	Admin does not mark Paid; order remains Payment Reported until resolved
Payment	UPI details not set by admin	Payment page shows notice; buyer told to contact shop
Image upload	File too large or wrong type	Rejected with message; product not saved
Admin	Admin edits product while a buyer has it in cart	Cart uses the latest price at checkout; buyer sees updated total before placing order
Backup/restore	Application restarted unexpectedly	Previously saved data intact; sessions require login again

11. Key User Flows

Flow 1 - Farmer First Use
1. Open home page > tap Farmer Management: Sign Up.
2. Fill registration form > submit.
3. Decision: valid? No - correct highlighted fields and submit again. Yes - go to Login.
4. Log in with email and password > Farmer Home.
5. Tap Crops > Add crop > save > see crop in list.
6. Tap Expenses > add first expense for that crop > save. Activation complete.

Flow 2 - Daily Expense Entry
1. Farmer Home > Expenses.
2. Enter details, date, choose crop, enter amount > Add.
3. Decision: valid? No - message beside field. Yes - "Saved" message and entry in list.
4. Optional: Edit or delete a wrong entry (with confirmation).

Flow 3 - Worker Payment Check
1. Farmer Home > Workers > add worker with days and daily wage.
2. Later: Workers > edit worker > update days worked.
3. Reports > Worker Payments > view payment amounts and grand total.

Flow 4 - Reminder
1. Farmer Home > Reminders > set time, date, description > Set.
2. On the reminder date, Farmer Home shows highlighted banner.
3. Farmer completes the task > deletes the reminder.

Flow 5 - Buyer Purchase
1. Open Mart > browse products.
2. Tap Add To Cart > Decision: logged in? No - login or register, then return. Yes - item added.
3. Open Cart > adjust quantity > Checkout.
4. Confirm delivery address > Place order > bill number created, status Pending Payment.
5. Payment page > pay via own UPI app > tap "I have paid" > status Payment Reported.
6. Admin verifies payment > marks Paid > buyer sees status Paid in My Orders.

Flow 6 - Admin Product Management
1. Admin login > Admin Home > Add Product.
2. Fill name, description, category, price, quantity, image > Save.
3. Product visible in Mart. Later: Products list > Edit or Delete (with confirmation).

Flow 7 - Admin Order Handling
1. Admin Home > Orders > filter by status Payment Reported.
2. Check own UPI app for the payment.
3. Decision: payment received? Yes - Mark Paid. No - leave status or Cancel the order (stock returned).

Failure Flow - Save Failure
1. Farmer submits form > network fails.
2. Page shows "Could not save. Please check internet and try again" with form values kept.
3. Farmer taps Add again > one record saved > "Saved" message.

Failure Flow - Out of Stock at Checkout

1. Buyer taps Place Order > stock check fails for one item.
2. No order created; message names the product and available quantity.
3. Buyer changes quantity in cart > Place Order again > order created.

12. Non-Functional Requirements

ID	Category	Requirement	Target / Acceptance
NFR-01	Performance	Page load for normal pages	Under 3 seconds on a typical 4G phone connection; page size under 300 KB excluding product images
NFR-02	Performance	Server response for save and list actions	95th percentile under 500 ms with 50 concurrent users
NFR-03	Performance	Report generation	Under 2 seconds for one farmer with 5,000 expense records
NFR-04	Scalability	Concurrent users supported by one server	100 simultaneous users without errors in a load test
NFR-05	Availability	Uptime of the running system	99% per month (single instance; planned restarts excluded)
NFR-06	Reliability	Data saved after confirmation message	Zero loss of saved records after a normal restart
NFR-07	Reliability	Duplicate prevention	Double-submit creates one record (FR-EXP-05 applies to all create forms)
NFR-08	Reliability	Order and stock consistency	Order creation and stock reduction succeed or fail together; stock never below zero
NFR-09	Reliability	Backup	Copying the data folder gives a complete backup; restore tested once before launch
NFR-10	Security	Password storage	Passwords stored only as salted, slow hashes; never in plain text or logs
NFR-11	Security	Transport	HTTPS supported and enabled in the recommended setup; session cookie marked secure and HttpOnly
NFR-12	Security	Input handling (OWASP)	All database access uses parameterised queries; all displayed user text is escaped; forms protected against cross-site request forgery; verified by a basic security checklist
NFR-13	Security	Authorization	Every server request checks role and record owner; verified by tests for farmer, buyer, admin and visitor
NFR-14	Security	Rate limiting	Login limited as per FR-AUTH-06
NFR-15	Security	File upload	Only JPG/PNG up to 1 MB; stored under a random file name; never executed
NFR-16	Privacy	Personal data	Only data in this PRD is collected; a short privacy notice is shown at registration; no data shared with third parties
NFR-17	Privacy	Deletion and retention	Admin can remove a user on request (process documented); retention period to be decided (Q-09)
NFR-18	Accessibility	Readability and touch	Base text at least 16 px; buttons at least 48 px high; contrast ratio at least 4.5:1 (WCAG AA)
NFR-19	Accessibility	Keyboard and screen reader	All actions possible by keyboard; every field has a label; every icon has text beside it
NFR-20	Compatibility	Browsers	Latest two versions of Chrome, Edge, Firefox, Safari; Chrome on Android 8 or later
NFR-21	Compatibility	Screen sizes	Works from 360 px phone width to desktop with no sideways scrolling
NFR-22	Localization	Format and language readiness	Rs. amounts, dd-mm-yyyy dates; all screen text kept in one place so Marathi or Hindi can be added later without code changes
NFR-23	Observability	Logging and monitoring	Application log file with daily rotation; errors recorded with time and request; a /health address returns OK when running
NFR-24	Resilience	Slow or weak network	Clear message on failure; entered form values preserved; no duplicate saves on retry
NFR-25	Resilience	Restart	Application starts automatically after the server reboots (documented one-line setup)
NFR-26	Deployability	Single package	One JAR file contains server and HTML pages; run with one command: java -jar kisankhata.jar

ID	Category	Requirement	Target / Acceptance
NFR-27	Deployability	No extra installs	Only a Java runtime is needed; no separate database, web server or application server
NFR-28	Deployability	Configuration	One settings file with port, admin email and password, data folder, shop name, contact; sensible defaults for all except admin password
NFR-29	Deployability	Startup and docs	Starts in under 15 seconds; deployment guide of one page; a non-expert completes setup in 30 minutes or less (KPI-09)
NFR-30	Deployability	Upgrade	Replace the JAR and restart; existing data is kept

13. Information Architecture and Key Screens

Navigation

Public: Home | About | Contact | Farmer Login / Sign Up | Mart Login / Sign Up / Visit Mart
Farmer: Farmer Home > Crops | Expenses | Workers | Water | Fertilizer/Pesticide | Reminders | Reports (Date-wise, Crop-wise, Worker Payments) | Help | Log out
Mart (Buyer): Mart Home > Products | Cart > Checkout > Payment > Bill | My Orders | Log out
Admin: Admin Home > Products (list, add, edit) | Users | Orders | Payment Settings | Log out

UI Principles (Simple for Farmers)

- One task per screen; a maximum of 6 fields per form; Add and Show actions side by side.
- Large buttons with a word label and a simple icon; high contrast; plain words ("Save", "Delete").
- Always confirm saves (green "Saved") and deletions (confirm box).
- Pick-lists and date pickers instead of typing wherever possible; mobile keyboard type matches the field (number, phone, email).
- Header on every page: Home, Back, Log out. No hidden menus.

Key Screens

Screen	User	Purpose	Primary Actions
Home	Visitor	Entry to Farmer and Mart areas	Login, Sign Up, Visit Mart
Farmer Sign Up / Login	Visitor, Farmer	Create account / enter	Sign Up, Login
Farmer Home	Farmer	Menu and today's reminders	Open module, Log out
Crops	Farmer	Add and list crops	Add Crop, Show Crops, Delete
Expenses	Farmer	Add, list, edit expenses	Add, Update, Show, Delete
Workers	Farmer	Add, list, edit workers	Add, Update, Show, Delete
Water	Farmer	Record watering	Save, Show, Delete
Fertilizer/Pesticide	Farmer	Record input use	Save, Show, Delete
Reminders	Farmer	Set and list reminders	Set, Show, Delete
Reports	Farmer	Choose report	Date-wise, Crop-wise, Worker Payments
Report result pages (3)	Farmer	View totals	Print, Back
Help	Farmer	Plain-language steps	Read
Mart Home / Products	Visitor, Buyer	Browse products	Add To Cart
Buyer Sign Up / Login	Visitor, Buyer	Create account / enter	Sign Up, Login
Cart	Buyer	Review items and total	Update, Remove, Checkout, Continue
Checkout	Buyer	Confirm delivery details	Place Order
Payment	Buyer	Show UPI details and total	I have paid, Cancel
Bill / My Orders	Buyer	See bill and order status	View, Cancel (if pending)
Admin Login	Admin	Enter admin area	Login
Admin Home	Admin	Counts and menu	Open section
Admin Products / Add / Edit	Admin	Manage catalogue	Add, Edit, Delete
Admin Users	Admin	View farmers and buyers	View, Deactivate (P1)
Admin Orders	Admin	Check and update orders	Filter, Mark Paid, Cancel
Payment Settings	Admin	Set UPI ID and QR	Save

Standard Screen States

State	Behaviour
Loading	Simple "Please wait" text; buttons disabled while saving
Empty	Friendly sentence plus the next action (for example "No crops added yet. Add your first crop.")
Error	Plain message with what to do next; never technical text
Success	Green "Saved" message on the same screen
Unauthorized	Redirect to the correct login page or "Not allowed" message
Offline / slow network	Message "Could not connect. Please check internet and try again"; values kept

14. High-Level Data Model

Conceptual only. No schema, indexes or code are defined here (those belong to the engineering design).

Farmer > Crop > Expense
Farmer > Worker
Farmer > Water Record
Farmer > Fertilizer/Pesticide Record
Farmer > Reminder
Buyer > Cart Item > Product
Buyer > Order > Order Item > Product (name and price copied at order time)
Admin > Product, Payment Settings

Entity	Purpose	Key Relationships	Owner	Lifecycle
Farmer	Account of a farmer	Owns crops, expenses, workers, records, reminders	Farmer	Registered > Active > Disabled (by admin)
Crop	A crop the farmer grows (name, category, plantation date, duration, description)	Belongs to one farmer; has many expenses	Farmer	Added > Edited > Deleted (blocked if expenses)
Expense	Money spent (details, date, amount)	Belongs to one farmer and one crop	Farmer	Added > Edited > Deleted
Worker	Person paid by the farmer (name, contact, days, daily wage, optional email/address/UPI)	Belongs to one farmer	Farmer	Added > Updated > Deleted
Water Record	Watering entry (plot, fertilizer yes/no, quantity, unit, date)	Belongs to one farmer	Farmer	Added > Deleted
Fertilizer/Pesticide Record	Input use entry (plot, product, quantity, unit, date)	Belongs to one farmer	Farmer	Added > Deleted
Reminder	Time, date, description	Belongs to one farmer	Farmer	Set > Due > Deleted
Buyer	Account of a buyer (with shipping address)	Owns cart items and orders	Buyer	Registered > Active > Disabled
Product	Item for sale (name, description, category, price, stock, image)	Created by admin; appears in cart items and order items	Admin	Added > Edited > Deleted (hidden)
Cart Item	Product and quantity chosen by a buyer	Belongs to one buyer and one product	Buyer	Added > Changed > Removed or converted to order
Order (Bill)	Bill number, date, delivery details, total, status	Belongs to one buyer; has many order items	Buyer / Admin	Pending Payment > Payment Reported > Paid, or Cancelled
Order Item	Product name, price and quantity as ordered	Belongs to one order	System	Created with order
Admin	Account of the shop operator	Manages products, orders, settings	Admin	Created from setup file
Payment Settings	UPI ID, QR image, shop contact	One set for the whole mart	Admin	Set > Updated

15. Roles and Permissions

Roles: **Visitor** (not logged in), **Farmer**, **Buyer**, **Admin**. C = Create, R = Read, U = Update, D = Delete.

Role	Resource / Module	Permission
Visitor	Home, About, Contact, Product list	R
Visitor	Farmer and Buyer registration, login	C (register), login
Farmer	Own profile	R
Farmer	Own crops, expenses, workers, water, fertilizer/pesticide, reminders	C, R, U (crops P1; others P0), D
Farmer	Own reports	R, print
Farmer	Other farmers' data, mart admin, any admin page	None
Buyer	Own profile and cart	R, U (cart), D (cart items)
Buyer	Own orders	C (place), R, cancel while Pending Payment (P1)
Buyer	Farmer module, admin pages, other buyers' orders	None
Admin	Products	C, R, U, D
Admin	Farmer and Buyer lists	R; deactivate/reactivate (P1) - sensitive
Admin	Orders	R, mark Paid, cancel - sensitive
Admin	Payment settings	R, U - sensitive
Admin	Farmers' crop, expense and worker records	None (not required for any admin task)

Sensitive operations (all written to the audit log, Section 18): marking an order Paid or Cancelled, changing payment settings, deactivating a user, deleting a product, admin login.

16. Integrations and Third-Party Services

Release 1 deliberately has almost no external dependencies, to keep deployment simple and cost low. No provider is named because none has been specified.

Integration	Purpose	Criticality	Failure Impact
UPI payment (static QR / ID shown on page)	Buyer pays in own UPI app; no technical connection to any bank or gateway	High for the mart	If details are missing or wrong, buyers cannot pay; mitigated by Payment Settings notice and admin check
Local file storage (server disk)	Product images and payment QR image	Medium	Images missing; the rest of the system keeps working
Browser print	Print-friendly reports	Low	None to core use
HTTPS certificate (hosting setup)	Secure access	High for security	Browser warnings; unencrypted passwords if skipped
Analytics, SMS, email, maps, KYC, push	Not used in release 1	n/a	n/a

17. Analytics and Instrumentation

Events are stored by the application itself (no external analytics service) and are used only to compute the KPIs in Section 5.

Event Name	Trigger	Key Properties	Purpose
farmer_registered	Farmer account created	date	KPI-01
farmer_login	Successful farmer login	date	Retention
crop_added	Crop saved	farmer, date	KPI-02
expense_added	Expense saved	farmer, date	KPI-02, KPI-03
worker_added	Worker saved	farmer, date	KPI-03
water_recorded / input_recorded	Water or fertilizer/pesticide saved	farmer, date	KPI-03
reminder_set	Reminder saved	farmer, date	KPI-03
report_viewed	Any report opened	report type	KPI-05
form_error	Validation error shown	form name, field	Guardrail (usability)
buyer_registered	Buyer account created	date	AARRR acquisition
cart_item_added	Item added to cart	product, quantity	KPI-06
order_placed	Order created	bill number, total	KPI-06
payment_reported	Buyer taps "I have paid"	bill number	Order funnel
order_paid	Admin marks Paid	bill number, total	KPI-07

Farmer funnel: Home > Sign Up > Login > First Crop > First Expense > Weekly Record.

Mart funnel: Mart Home > Add To Cart > Checkout > Payment Reported > Paid.

18. Audit and Operational Logging

Action	Who	What is recorded	Reason required?
Admin login (success and failure)	Admin	Time, email, result	No
Farmer / buyer failed logins	System	Time, email, count	No
Product add / edit / delete	Admin	Time, admin, product, old and new values	No
Order marked Paid / Cancelled	Admin	Time, admin, bill number, old and new status	Yes for Cancel
Payment settings changed	Admin	Time, admin, field changed (not full values of secrets)	No
User deactivated / reactivated	Admin	Time, admin, user	Yes
User data removal on request	Admin	Time, admin, user	Yes
Application errors	System	Time, request, error	n/a

Audit records cannot be edited from the website and are visible only to the admin or the server operator.

19. Release Plan and Roadmap

Phase	Theme	Key Scope	Exit Criteria
Phase 0	Foundations	Project setup, single-JAR build, configuration file, embedded database, base page layout, login and session handling, deployment guide draft	JAR starts with one command; empty home page and health address work; security checklist items for sessions and passwords pass
Phase 1	Farmer Core	Farmer registration/login, Farmer Home, crops, expenses, workers with payment amount, three reports	All Phase 1 P0 requirements pass acceptance; 5 farmers complete add-expense task in 30 seconds or less
Phase 2	Farm Operations	Water, fertilizer/pesticide, reminders and banner	All related P0 requirements pass; soft-launch checkpoint reached (farmer-only release possible)
Phase 3	Mart and Admin (MVP release)	Buyer flow, cart, checkout, payment page, bill; admin products, users, orders, payment settings	All P0 requirements pass; end-to-end test of an order from cart to Paid; 30-minute deployment test passed; MVP complete
Phase 4	Polish (fast-follow)	All P1: edit crop, duplicate check, login block, print view, help, About/Contact, My Orders, buyer cancel, user deactivate, admin counts	P1 acceptance passed; pilot feedback reviewed
Future	Ideas (P2, not planned)	Translations, notifications, richer reports	Decided after pilot data

Dates are not given because budget and team size are unknown (Q-11). Phases are ordered by dependency and value.

20. MVP Definition

The MVP is complete when a new farmer can register, record crops, expenses, workers, water, fertilizer/pesticide and reminders and view all three reports; a buyer can order a product, pay by UPI and receive a bill; an admin can manage products, see users and orders and mark payments; and one person can deploy the whole system from a single JAR in 30 minutes or less.

Must Have (P0)

- Farmer: registration, login/logout, session handling, Farmer Home (FR-HOME-01, 03; FR-AUTH-01 to 05).
- Crops (add, list, delete), expenses (add, list, edit, delete), workers (all) and payment amount.
- Water and fertilizer/pesticide records; reminders including home banner.
- Date-wise, crop-wise and worker-wise reports.
- Mart: browse, buyer sign-up/login, cart, checkout with stock check, payment page, bill.
- Admin: login, product add/list/edit/delete, user lists, orders report, mark Paid, payment settings.
- All Section 12 NFRs for security, reliability and deployability.

Should Have (P1)

- Edit crop, duplicate crop check, login attempt limit, print view, Help, About/Contact, My Orders, buyer cancel, user deactivation, admin counts.

Later (P2)

- Translations (Marathi, Hindi), notifications outside the website, richer reports.

21. Risks and Mitigations

Risk	Probability	Impact	Mitigation
Farmers find screens confusing or too slow to use (adoption)	Medium	High	Few fields, big buttons, plain words; test with 5 farmers before launch; KPI-08 gate
Wrong or missing data entered, making reports misleading	Medium	Medium	Validation, confirmation messages, edit and delete allowed, totals shown
Buyer taps "I have paid" without paying, or admin marks wrong order Paid	Medium	High	Admin verifies in own UPI app using bill number; audit log of every status change
Data loss because data is a single local file	Low	High	Documented backup by copying data folder; restore tested before launch (NFR-09)
Passwords or personal data exposed (no HTTPS, weak handling)	Low	High	Hashed passwords, HTTPS in setup guide, parameterised queries, security checklist (NFR-10 to 15)
Wrong Java version on host prevents start	Medium	Medium	State Java 17+ clearly; start-up check prints a clear message
Server restart or crash stops the service	Medium	Medium	Auto-start setup (NFR-25); health address for monitoring
Stock goes negative when two buyers order together	Low	Medium	Order and stock update done together (NFR-08); concurrent-order test
Scope creep with extra features	High	Medium	Non-goals list; any new feature needs Product Owner approval and a PRD change
Marketplace has no products or no buyers (adoption)	Medium	Medium	Admin loads initial products before launch; farmer module can launch alone (soft-launch checkpoint)
Uploaded image abuse or oversized files	Low	Low	Type and size limits; random file names (NFR-15)
Legal or privacy obligations not identified	Medium	Medium	Ask a qualified advisor before launch (Q-10); collect only necessary data

22. Open Questions

#	Question	Owner	Needed By
Q-01	Is "KisanKhata" the final product name, and what are the names of the Farmer area and the Mart?	Business / Founder	Before design
Q-02	Which embedded database will Engineering use, and is it acceptable to the operator?	Engineering Lead	Phase 0
Q-03	Confirm the single-seller mart model (admin sells all products) and what products are sold.	Business / Founder	Phase 0
Q-04	Is a daily wage field acceptable for workers (needed for payment amount)?	Product Manager	Phase 1
Q-05	Which language(s) should the first release use: English only, or Marathi too?	Business / Domain advisor	Before design
Q-06	Who provides the UPI ID/QR, and who verifies payments daily?	Business / Founder	Phase 3
Q-07	Cancellation and refund policy for the mart (not covered by the system).	Business / Founder	Phase 3
Q-08	Delivery charges or delivery area limits? (none assumed)	Business / Founder	Phase 3
Q-09	How long is user data kept and what is the process for deletion requests?	Operations Lead	Before launch
Q-10	Which privacy or consumer-protection requirements apply? (to be confirmed by a qualified advisor)	Business / Founder	Before launch
Q-11	Team size, budget and target launch date.	Business / Founder	Phase 0
Q-12	Where will the system be hosted, and who owns the domain name?	Operations Lead	Phase 3
Q-13	Should crop name be a free text field or a fixed list of common crops?	Domain advisor	Phase 1

23. Traceability Check

Chain checked: Business Goal > Product Goal > KPI > User Story > Functional Requirement > Acceptance Criteria > Flow > Phase.

Business Goal	Product Goal	KPI	Main Requirements	Flow	Phase
BG-01	PG-01, PG-04	KPI-01, 02, 08	FR-AUTH-01 to 05, FR-HOME-03, FR-CROP-01	1, 2	1
BG-02	PG-01, 02, 03	KPI-03, 04, 05	FR-EXP-01 to 05, FR-WRK-01 to 05, FR-WTR, FR-PST, FR-REM, FR-RPT-01 to 03	2, 3, 4	1, 2
BG-03	PG-05	KPI-06, 07	FR-MART-01 to 09, FR-ADM-03 to 07, 09 to 11	5, 6, 7	3
BG-04	PG-06	KPI-09	NFR-26 to 30	-	0, 3
BG-05	PG-06	-	NFR-27, Constraint "no paid services"	-	0

Gaps found and resolved

- Worker report had an empty payment column in the original screens: resolved by FR-WRK-05 and assumption A-05.
- Admin sign-up existed in the original report: replaced by setup-file admin (A-03) for safety; flagged for confirmation.
- Payment existed only as a static QR with Done/Cancel: made testable through order statuses (FR-MART-08, FR-ADM-10).
- Every P0 requirement has acceptance criteria; every P0 requirement sits in Phases 1 to 3, all before the MVP release.
- BG-05 has no KPI: accepted as a constraint check rather than a measured KPI.
- Every user story in Section 7 maps to at least one functional requirement.

24. Glossary

Term	Definition
KisanKhata	Product name. "Kisan" means farmer; "Khata" means ledger or account book
Farmer Management	The part of the website where farmers keep farm records
KisanKhata Mart	The part of the website where buyers order products from the admin
Admin	The shop operator who manages products, users, orders and payment settings
Plot	A named or numbered part of the farmland used in water and fertilizer/pesticide records
Fertilizer / Pesticide	Substances applied to crops, recorded with name, quantity and date
Bill number	Unique number given to every order
Order status	Pending Payment, Payment Reported, Paid, Cancelled
UPI	Unified Payments Interface, the Indian instant payment system
QR	Square code scanned in a UPI app to pay
JAR	Single Java file that contains the whole running application
PRD	Product Requirements Document
MVP	Minimum Viable Product: smallest complete release that delivers the core value
NFR	Non-Functional Requirement (quality such as speed, security, accessibility)
KPI	Key Performance Indicator
WAF	Weekly Active Farmers, the North Star Metric
OWASP	Open Web Application Security Project, a widely used list of web security risks
WCAG	Web Content Accessibility Guidelines
HTTPS	Secure version of web traffic